

# CORE 1

## Motion in 2D - Projectile Motion

Orientation: canonical Motion in 2D first, then the five explicitly labelled demand extensions.

### Purpose

Build the map before calculation. This Core is a compact orientation, not the full lesson.

## 1. Scope and conventions

- Use one declared Cartesian x-y frame and one shared physical clock.
- Apply constant-acceleration relations per component only when the relevant acceleration component is constant.
- Select the projectile model from the retained forces before using projectile shortcuts.

## 2. Canonical spine

### R1 - Independent x/y components, one shared clock

Perpendicular component equations can be solved separately, but they describe one object at one physical time.

#### Exit check

A particle has  $v_x=6$  m/s and  $v_y=-8$  m/s at  $t=3$  s. Explain what 'independent components' means and what must remain shared.

### R2 - Constant-acceleration kinematics component by component

The one-dimensional constant-acceleration equations apply independently to x and y only when each acceleration component is constant over the interval.

#### Exit check

A particle starts with  $u=(0,5)$  m/s and constant  $a=(10,4)$  m/s<sup>2</sup>. Set up x and y displacement equations at common time t.

### R3 - Select the gravity-only projectile model before using projectile consequences

Ideal projectile motion is ordinary 2D component kinematics with  $a_x=0$  and  $a_y=-g$  (for +y upward), selected only after retained forces are checked.

#### Exit check

A stone is launched horizontally from a cliff with speed u. Write its component acceleration, velocity, and displacement relations.

## 3. Demand-driven extension map

### E1 - EXTENSION: instantaneous velocity direction from $x(t)$ , $y(t)$

Direction of motion comes from tangent velocity ( $dx/dt$ ,  $dy/dt$ ), not the position vector (x,y).

### E2 - EXTENSION: projectile rebound after an energy-changing impact

A kinetic-energy retention fraction multiplies  $v^2$ . After impact creates a new launch state, the new vertical component controls the next rise height.

### E3 - EXTENSION: projectile motion when gravity changes after the apex

A change in constant gravity creates two stages joined by one boundary state. Position and velocity are inherited; acceleration changes.

### E4 - EXTENSION: trajectory equation through a specified point

Eliminating the shared time gives a path equation  $y(x)$ , but point membership and branch/apex location are separate decisions.

### E5 - EXTENSION: projectile motion with linear drag

Component splitting survives, but the constant-acceleration law fails where acceleration depends on velocity.

## 4. Before moving on

#### Self-check

Can you explain why  $x$  and  $y$  may be solved separately but cannot use different times for the same state?

#### Self-check

Can you identify when the standard projectile model is invalid because an impact, force change, or drag law modifies the dynamics?